

PE-EC801B — Fiber Optic Communication

Semester VIII • 3L:0T:0P • 3 Credits

Module 4 Exam Notes

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Optical Switches

Optical Switch

An **optical switch** is a device that selectively routes an optical signal from one port to another without converting it to an electrical signal. Switches are essential for optical cross-connects (OXCs), protection/restoration, and reconfigurable networks.

1.1 Classification of Optical Switches

Type	Mechanism	Switching Time	Remarks
Mechanical	Physical fiber/mirror movement	ms to s	Low loss; no polarization dep.; slow
Electro-optic	Pockels effect in LiNbO_3	ps to ns	Fast; polarization dependent; insertion loss
Acousto-optic	Acousto-optic diffraction (Bragg)	μs	Can switch wavelengths; tunable
Thermo-optic	Thermo-optic effect in waveguide	ms	Low cost (Si/polymer); compact
MEMS	Micro-mirror arrays (MEMs)	ms	Large port count; low loss

2. Directional Couplers — Coupled Mode Analysis

Directional Coupler

A **directional coupler** is a 4-port device formed by bringing two waveguides or fibers into close proximity so that their evanescent fields overlap. Power is transferred from one waveguide to the other via evanescent coupling over the coupling length.

2.1 Coupled Mode Equations

Let $a_1(z)$ and $a_2(z)$ be the field amplitudes in waveguides 1 and 2. Coupled mode theory gives:

$$\frac{da_1}{dz} = -j\kappa a_2 e^{j\Delta\beta z}, \quad \frac{da_2}{dz} = -j\kappa a_1 e^{-j\Delta\beta z}$$

where κ is the coupling coefficient (dependent on overlap integral of evanescent fields) and $\Delta\beta = \beta_1 - \beta_2$ is the phase mismatch.

Power Transfer in a Directional Coupler

For identical waveguides ($\Delta\beta = 0$, perfect phase matching), the power in each arm as a function of length L :

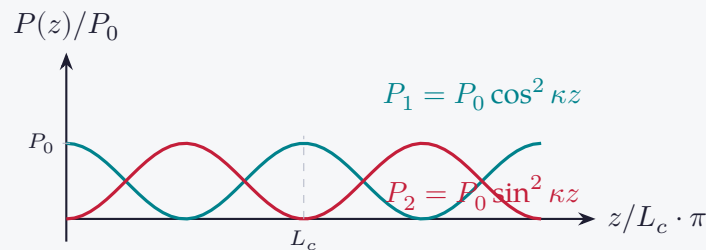
$$P_1(L) = P_0 \cos^2(\kappa L), \quad P_2(L) = P_0 \sin^2(\kappa L)$$

Coupling length: $L_c = \pi/(2\kappa)$ — full power transfer from waveguide 1 to 2.

For partial coupling ratio ϕ : $L = \phi\pi/(2\kappa)$.

With phase mismatch: $\max(P_2/P_0) = \kappa^2/(\kappa^2 + (\Delta\beta/2)^2)$.

Directional Coupler — Power Transfer Along Coupling Length



3. Mach-Zehnder Modulator (MZM)

Mach-Zehnder Modulator (MZM)

A **Mach-Zehnder Modulator** is an electro-optic intensity modulator based on LiNbO_3 . Light is split into two arms of a Mach-Zehnder interferometer; an applied electric field via the Pockels effect changes the refractive index of one (or both) arms, causing a phase shift. When the two beams recombine, constructive or destructive interference produces intensity modulation.

3.1 Structure and Operation

1. Input light is split 50:50 at a Y-junction.
2. Electrodes on each arm apply voltage V across the waveguide; the Pockels effect changes the refractive index $\Delta n \propto r_{33}E$ where r_{33} is the electro-optic coefficient ($\approx 30 \text{ pm/V}$ for LiNbO_3).
3. A phase shift $\Delta\phi = \pi V/V_\pi$ is induced.
4. The two arms recombine at a second Y-junction.

MZM Transfer Function

$$P_{out}(V) = P_{in} \cos^2\left(\frac{\pi V}{2V_\pi}\right) = \frac{P_{in}}{2} \left[1 + \cos\left(\frac{\pi V}{V_\pi}\right) \right]$$

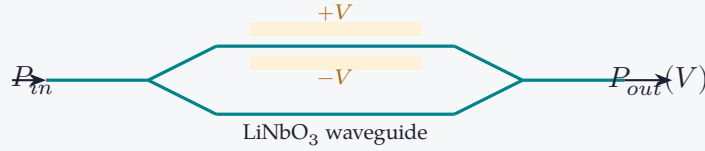
Half-wave voltage V_π : The voltage required to shift the phase by π (switches from max

to min transmission).

$$\text{Extinction ratio (ER): } r_{ext} = P_{max}/P_{min} \quad (\text{dB}) = 10 \log_{10}(r_{ext})$$

Ideal MZM: $r_{ext} \rightarrow \infty$ (perfect extinction). Practical: 20–30 dB.

Mach-Zehnder Modulator — Structure and Output



4. EDFA — Erbium-Doped Fiber Amplifier

EDFA

An **Erbium-Doped Fiber Amplifier (EDFA)** is an optical amplifier that amplifies light directly in the optical domain by stimulated emission from excited Er^{3+} ions. It operates in the 1530–1565 nm C-band (or 1570–1610 nm L-band), exactly matching the lowest-loss window of silica fiber.

4.1 Structure and Pumping

An EDFA consists of a few meters of silica fiber doped with Er^{3+} ions. A pump laser excites the erbium ions to higher energy levels. Two standard pump wavelengths:

- **980 nm pump:** Three-level pumping; low noise figure ($\sim 3\text{--}4$ dB); requires WDM coupler.
- **1480 nm pump:** Two-level pumping (directly pumps to metastable $^4I_{13/2}$ level); higher output power; noise figure $\sim 4\text{--}5$ dB.

4.2 Gain Spectrum and ASE Noise

EDFA provides gain over ~ 35 nm bandwidth (1530–1565 nm). The gain spectrum is not flat; tilt compensation requires gain flattening filters (GFF).

Amplified Spontaneous Emission (ASE): Not all spontaneous photons escape the fiber; a fraction is guided along the fiber and amplified, producing broadband noise. The ASE power spectral density in each polarization is:

$$S_{ASE}(\nu) = n_{sp} h\nu (G - 1)$$

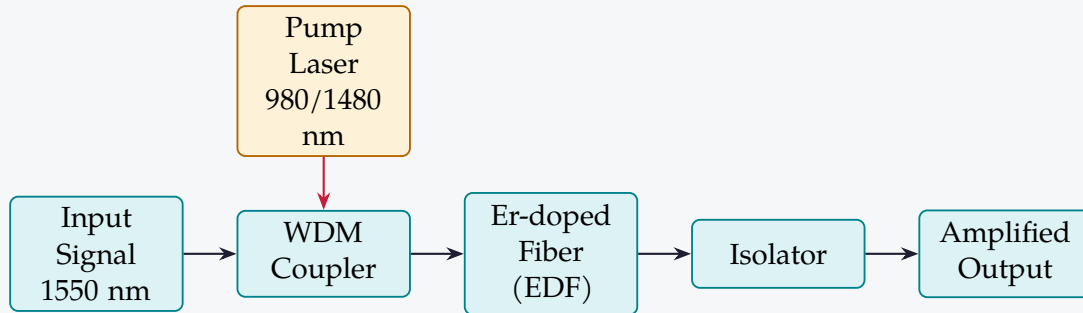
where $n_{sp} \geq 1$ is the spontaneous emission factor ($n_{sp} = N_2/(N_2 - N_1)$ for a two-level system).

Noise Figure (NF) of an EDFA:

$$\text{NF} = 2n_{sp} \frac{G - 1}{G} + \frac{1}{G} \approx 2n_{sp} \quad \text{for large } G$$

Minimum theoretical NF = 3 dB (for $n_{sp} = 1$, ideal inversion).

EDFA Structure — Forward Pumped Configuration



5. Raman Amplifier

Raman Amplifier

A **Raman amplifier** exploits **Stimulated Raman Scattering (SRS)** in silica fiber to amplify an optical signal. A high-power pump transfers energy to a signal that is Stokes-shifted by ~ 13 THz (corresponding to ~ 100 nm red shift). The transmission fiber itself acts as the gain medium.

5.1 Stimulated Raman Scattering (SRS)

A pump photon at frequency ν_p interacts with a silica molecular vibration to produce a signal photon at lower frequency $\nu_s = \nu_p - \nu_v$ (Stokes shift) and a phonon. The Raman gain coefficient $g_R \approx 6 \times 10^{-14}$ m/W at $\lambda_p = 1450$ nm.

Raman Gain

$$G_R = \exp\left(\frac{g_R P_p L_{eff}}{A_{eff}}\right)$$

where P_p is pump power, $L_{eff} = [1 - e^{-\alpha_p L}]/\alpha_p$ is the effective length, A_{eff} is effective mode area.

5.2 Distributed vs Lumped Raman Amplification

- **Distributed Raman:** The transmission fiber is used as the gain medium with backward pumping. The signal is amplified continuously along the fiber, reducing the noise figure dramatically. Best OSNR performance.
- **Lumped (discrete) Raman:** A separate short high-nonlinearity fiber (HNF) provides concentrated gain. Simpler pump management.

5.3 EDFA vs Raman Amplifier Comparison

Parameter	EDFA	Raman Amplifier
Gain medium	Er-doped fiber	Transmission fiber (silica)
Gain band	1530–1565 nm (C-band)	Flexible (~ 100 nm from pump)
Pump wavelength	980/1480 nm	~ 100 nm below signal
Gain flatness	Requires GFF	Inherently broad but uneven
Noise figure	~ 3 – 6 dB	< 0 dB (effective; distributed)
Amplification type	Lumped	Distributed or lumped
Cost/complexity	Moderate	Higher pump power needed

Quick Revision — Module 4

Optical Switches, Couplers, and Amplifiers | PE-EC801B

Key Formulas at a Glance

Coupler power transfer	$P_1 = P_0 \cos^2(\kappa L); \quad P_2 = P_0 \sin^2(\kappa L)$
Coupling length	$L_c = \pi/(2\kappa)$ (full transfer)
MZM transfer function	$P_{out} = P_{in} \cos^2(\pi V/2V_\pi)$
EDFA ASE noise PSD	$S_{ASE} = n_{sp} h\nu (G - 1)$
EDFA noise figure	$NF \approx 2n_{sp}$ (large G); min = 3 dB
Raman gain	$G_R = \exp(g_R P_p L_{eff}/A_{eff})$
Raman Stokes shift	$\Delta\nu \approx 13$ THz; $\Delta\lambda \approx 100$ nm at 1550 nm

One-Line Definitions

- **Directional coupler:** 4-port device transferring power via evanescent field overlap between two waveguides.
- **Coupling coefficient κ :** Rate of power transfer per unit length in a directional coupler.
- **MZM:** Mach-Zehnder Modulator; electro-optic intensity modulator using LiNbO_3 Pockels effect.
- V_π : Voltage causing π phase shift in MZM; half-wave voltage.
- **EDFA:** Er-doped fiber amplifier; all-optical gain in 1530–1565 nm C-band.
- **ASE:** Amplified Spontaneous Emission; broadband noise generated in optical amplifiers.
- **SRS:** Stimulated Raman Scattering; pump photon excites molecular vibration, emits Stokes-shifted signal photon.
- **Distributed Raman:** Uses the transmission fiber as gain medium; best noise performance.
- **Noise figure (NF):** Degradation of OSNR through an amplifier; minimum theoretical = 3 dB.

Frequently Asked / Viva Questions

Q1. What is the principle of operation of a directional coupler?

When two waveguides are placed sufficiently close, their evanescent fields overlap. The resulting coupled mode equations show that power oscillates sinusoidally between the two guides with period $L_c = \pi/(2\kappa)$.

Q2. What is the coupling length and how is it related to κ ?

Frequently Asked / Viva Questions (contd.)

$L_c = \pi/(2\kappa)$ is the length at which all power transfers from guide 1 to guide 2. κ (m^{-1}) depends on the overlap integral of the evanescent fields.

Q3. What is the half-wave voltage V_π of an MZM?

V_π is the voltage needed to cause a phase shift of exactly π in one arm of the MZM, switching the output from maximum to minimum (or vice versa). Lower V_π means less drive power needed.

Q4. Why is LiNbO_3 used in MZMs?

LiNbO_3 has a large electro-optic (Pockels) coefficient $r_{33} \approx 30 \text{ pm/V}$, enabling efficient phase shifting with moderate voltage. It also has low optical loss in the 1310–1550 nm bands.

Q5. What are the two pump wavelengths for EDFA and their trade-offs?

980 nm: lower noise figure ($\sim 3\text{--}4 \text{ dB}$), three-level pumping, lower output power.
1480 nm: higher saturation power, two-level pumping, slightly higher NF ($\sim 4\text{--}5 \text{ dB}$). 980 nm preferred for low-noise preamplification.

Q6. What is ASE in EDFA? How does it affect system performance?

ASE is spontaneous emission from excited Er^{3+} ions that is guided and amplified. It degrades OSNR (optical signal-to-noise ratio) and limits the number of amplifier cascades in a chain.

Q7. What is the minimum theoretical noise figure of an EDFA?

3 dB. This corresponds to $n_{sp} = 1$ (complete population inversion) and large gain $G \gg 1$.

Q8. Explain stimulated Raman scattering in the context of fiber amplification.

A high-power pump photon (ν_p) interacts with a silica phonon, losing energy to produce a Stokes signal photon at $\nu_s = \nu_p - \nu_{\text{phonon}}$. If a signal is present at ν_s , it triggers more SRS events — stimulated Raman amplification.

Q9. What are the advantages of distributed Raman amplification?

The gain is spread over the entire span, keeping the signal power higher throughout the fiber. This reduces the effective noise figure and improves OSNR compared to lumped amplification. No discrete amplifier components needed in the span.

Q10. Classify optical switches and state their switching speeds.

Mechanical: ms–s; Electro-optic: ps–ns; Acousto-optic: μs ; Thermo-optic: ms; MEMS: ms.

Q11. What is the extinction ratio of an MZM?

$\text{ER} = P_{\text{max}}/P_{\text{min}}$. An ideal MZM has infinite ER (perfect extinction). Practical devices achieve 20–30 dB due to waveguide asymmetries and electrode misalignment.

EDFA vs Raman — Quick Comparison

Feature	EDFA	Raman
Gain band	1530–1565 nm	Tunable with pump
Noise figure	≥ 3 dB	< 0 dB effective (dist.)
Amplifier type	Lumped	Distributed or lumped
Gain medium	Er-doped fiber	Transmission fiber
Pump λ	980/1480 nm	~ 1450 nm (for 1550)